WORKSHEET 2.6

Derivatives of Sine, Cosine, e^x and $\ln x$

Unit 2: The Derivative

Key ideas

- $\dfrac{d}{dx}[\sin x] = \cos x$ and $\dfrac{d}{dx}[\cos x] = -\sin x$ (x in radians).
- $\dfrac{d}{dx}[e^x] = e^x$ and $\dfrac{d}{dx}[\ln x] = \dfrac{1}{x}$ for $x > 0$.
- These combine with the sum and constant-multiple rules exactly as powers do.
- Exact trigonometric values are expected: know sine and cosine at 0, $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$.

Worked examples

Example 1 · A sum

Find $\dfrac{d}{dx}[3\sin x - 4\cos x + 2e^x]$.

Solution. Differentiate each term; the derivative of $-4\cos x$ is $+4\sin x$.

$3\cos x + 4\sin x + 2e^x$.

Example 2 · A tangent line

Write an equation of the line tangent to $y = \ln x$ at $x = 2$.

Solution. The point is $(2, \ln 2)$ and the slope is $\dfrac{1}{2}$.

$y - \ln 2 = \dfrac{1}{2}(x - 2)$.

Example 3 · Horizontal tangents

Find the values of x in $[0, 2\pi]$ at which $y = \sqrt{3}\,x + 2\cos x$ has a horizontal tangent.

Solution. $y' = \sqrt{3} - 2\sin x = 0$ gives $\sin x = \dfrac{\sqrt{3}}{2}$.

$x = \dfrac{\pi}{3}$ **and** $x = \dfrac{2\pi}{3}$.

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Practice

CORE

1. Find $\frac{d}{dx}[5e^x]$.

2. Find $\frac{d}{dx}[2\sin x + x^3]$.

3. Find $\frac{d}{dx}[4\ln x - x]$.

4. For $f(x) = \cos x$, find $f'\left(\frac{\pi}{6}\right)$.

EXAM LEVEL

5. Write an equation of the line tangent to $y = 2\sin x$ at $x = \frac{\pi}{3}$.

6. At what value of x does the graph of $y = 2x - e^x$ have a horizontal tangent?

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7. Evaluate $\lim_{h \to 0} \dfrac{e^{3+h} - e^3}{h}$.

8. Let $f(x) = 6 \ln x + x^2$ for $x > 0$. Find all values of x at which $f'(x) = 7$.

Question 9 · Free-response practice

Let $f(x) = 4 \sin x + 2x$ for $0 \leq x \leq 2\pi$.

(a) Find $f'(x)$.

(b) Find all values of x in the interval at which the graph of f has a horizontal tangent.

(c) Write an equation of the line tangent to the graph of f at $x = 0$.

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Worked solutions

1. $5e^x$.
2. $2\cos x + 3x^2$.
3. $\dfrac{4}{x} - 1$.
4. $-\sin\dfrac{\pi}{6} = -\dfrac{1}{2}$.
5. The point is $\left(\dfrac{\pi}{3}, \sqrt{3}\right)$ and the slope is $2\cos\dfrac{\pi}{3} = 1$: $y - \sqrt{3} = x - \dfrac{\pi}{3}$.
6. $2 - e^x = 0$ at $x = \ln 2$.
7. The derivative of e^x at 3: e^3.
8. $\dfrac{6}{x} + 2x = 7$ gives $2x^2 - 7x + 6 = (2x - 3)(x - 2) = 0$: $x = \dfrac{3}{2}$ and $x = 2$.
9. (a) $f'(x) = 4\cos x + 2$.
   (b) $\cos x = -\dfrac{1}{2}$: $x = \dfrac{2\pi}{3}$ and $x = \dfrac{4\pi}{3}$.
   (c) $f(0) = 0$ and $f'(0) = 6$: $y = 6x$.

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